

DAS732 — Programming Assignment 1: Visual Exploration of Natural Disasters

Team Pythagoras

September 2026

1. Dataset Description

Dataset: EM-DAT Natural Disasters Emergency Events Database (Country Profiles) **Source:** kaggle.com/datasets/mexwell/natural-disasters-emergency-events-database **Scope:** 10,431 records, 1900–2023, 225 countries, 13 natural disaster types.

Fields: Year, Country, ISO, Disaster Group, Disaster Subgroup, Disaster Type, Disaster Subtype, Total Events, Total Affected, Total Deaths, Total Damage (USD, original), Total Damage (USD, CPI-adjusted), CPI.

Preprocessing:

- Converted delimiter from ; to , for tool compatibility.
- Fixed source typos (“Disaster Subroup” → “Disaster Subgroup”, trailing space on “Extreme temperature”).
- Corrected CPI field (was stored with comma decimal separators).
- Left Total Affected / Total Deaths / Damage fields as null where unreported — not imputed as zero, since “zero impact” and “not reported” are materially different in this dataset.
- Added a derived Reporting Era field (Pre-1980 vs 1980–present) to flag that pre-1980 records are undercounted due to sparser global disaster-reporting infrastructure at the time — a caveat for interpreting long-run trends.

Missingness: Total Affected (2,845 missing), Total Deaths (3,056), Damage original (6,597), Damage adjusted (6,601), Disaster Subtype (2,133, labeled “Not specified”).

2. Guiding Question

How have natural disasters evolved in frequency, geographic distribution, and human/economic impact over the last century, and which disaster types and regions demand the most attention today?

This question is addressed through three complementary, non-overlapping task sets — one per team member — each covering a cohesive subset of variables, per the assignment’s task-design guidance.

3. Tasks and Visualizations

Per TA/instructor guidance (Sep 10, 2026): up to 15 visualizations may be built per person, but only a curated subset — favoring multi-variable, informative charts over simple 2-variable ones, across a variety of chart types — should appear below. Every figure has a number, caption, in-text reference, and interpretation.

3.1 Task Set A — Temporal Trends (Arnav Jain)

Variables used: Year, Disaster Type / Subtype, Total Events, Reporting Era

Tasks:

- *Overview* — total disaster events per year/decade, 1900–2023.
- *Trends* — growth or decline in frequency by disaster type (e.g., storms/floods vs. earthquakes).
- *Comparison* — pre-1980 vs. 1980-present event rates, to account for the reporting-bias caveat.

Planned chart types: stacked area chart (events by type over time), line chart (yearly totals), small multiples per disaster type.

Figure 1. *[Chart title]* — image placeholder for `images/Fig1.png`. *Caption:* [One or two sentences describing what the figure shows and which variables/encodings are used.]

Interpretation: [What this figure reveals in answer to the guiding question; reference it in prose as “As shown in Figure 1, ...”]

Figure 2. *[repeat pattern for each figure selected for this task set — aim for ~4-6, prioritizing multi-variable charts]*

3.2 Task Set B — Geographic Distribution (Kalpit Phogat)

Variables used: Country, ISO, Disaster Type, Total Events

Tasks:

- *Overview* — choropleth map of total disaster events by country.
- *Search/filter* — which countries are most exposed to which disaster type (e.g., earthquakes in Japan/Turkey, floods in India/Bangladesh).
- *Comparison* — regional/continental disaster-type profiles.

Planned chart types: choropleth map, filterable bar chart by country, heatmap (country × disaster type).

Figure N. *[Chart title]* — image placeholder for `images/FigN.png`. *Caption:* [description + variables/encodings used.] *Interpretation:* [finding, referenced in prose as “Figure N shows ...”]

Figure N+1. *[repeat for ~4-6 figures for this task set]*

3.3 Task Set C — Impact Severity (Lakshya Jain)

Variables used: Total Deaths, Total Affected, Total Damage (USD, adjusted), CPI

Tasks:

- *Overview* — top-10 deadliest and costliest disaster types/countries.
- *Trends* — economic damage over time (CPI-adjusted to normalize for inflation).
- *Correlation* — relationship between event frequency and severity.

Planned chart types: ranked bar charts, scatterplot (events vs. deaths), dual-axis time series (damage vs. deaths).

Figure M. *[Chart title]* — image placeholder for images/FigM.png. *Caption:* [description + variables/encodings used.] *Interpretation:* [finding, referenced in prose as “Figure M shows ...”]

Figure M+1. *[repeat for ~4-6 figures for this task set]*

4. Overall Conclusions

[Synthesis across all three task sets — to be filled in once visualizations and inferences are complete. This section should tie Temporal, Geographic, and Impact findings back into a single data story answering the guiding question.]

5. Author Contributions

Member	Roll Number	Task Set	Contribution
Arnav Jain	BT2024233	A — Temporal Trends	<i>[to be filled]</i>
Kalpiti Phogat	BT2024093	B — Geographic Distribution	<i>[to be filled]</i>
Lakshya Jain	BT2024044	C — Impact Severity	<i>[to be filled]</i>

Data preprocessing: *[name of member(s) who handled cleaning — also opens the first minute of the video demo]*

Report writing/compilation: *[to be filled]*

6. AI Declaration

Separate AI declaration forms for each team member are included as an appendix (see ai_declaration_form.md / attached forms).